# Fatima Jinnah Women Univeristy



*Course Title : Web technology*

## *Topic: Tourism guide website*


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## 1. Project Overview

### 1.1 Introduction

The Pakistan Tourism Guide Website is a comprehensive web application designed to promote tourism in Pakistan. The platform allows users to explore tourist destinations, browse tour packages, book trips, and manage their travel plans. The system includes user authentication, profile management, shopping cart functionality, and an interactive interface for seamless user experience.

### 1.2 Project Objectives

The main objectives of this project are:
• To create a user-friendly platform for Pakistan tourism
• To implement secure user authentication and session management
• To provide online booking capabilities for tour packages
• To showcase Pakistani destinations with detailed information
• To build a responsive website that works on all devices

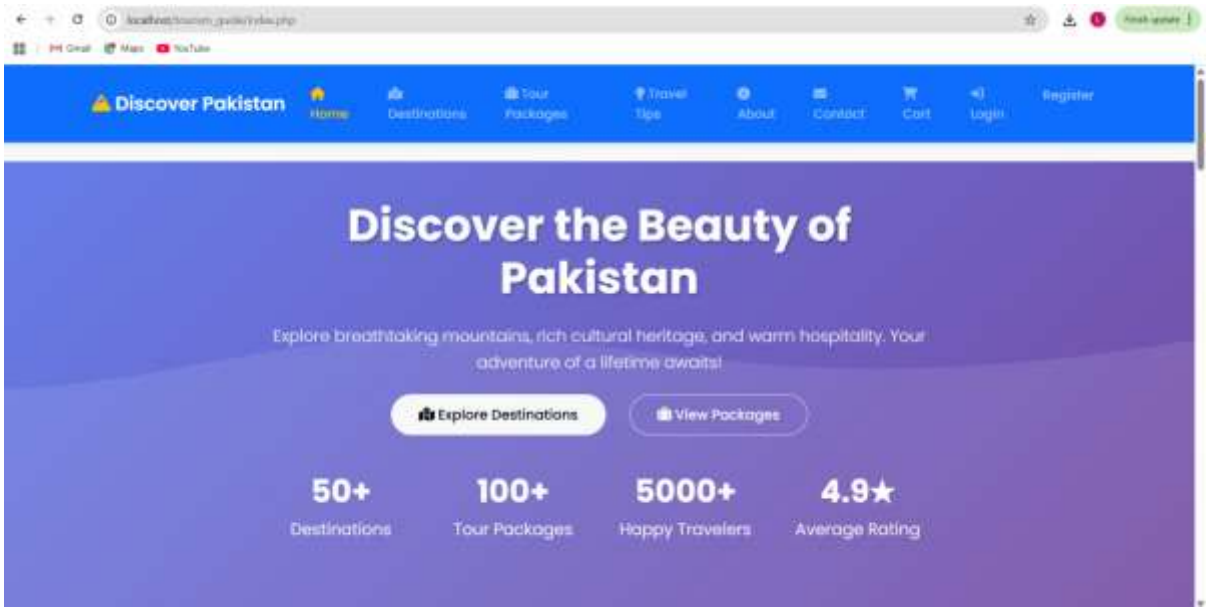



Figure 1: Homepage.

## 2. Technology Stack

### 2.1 Frontend Technologies

| Technology | Purpose |
|---|---|
| HTML | Page structure and content |
| CSS | Styling and layout design |
| JavaScript | Interactive features and form validation |

### 2.2 Backend Technologies

| Technology | Purpose |
|---|---|
| PHP | Server-side programming and business logic |
| MySQL | Database management system |
| Apache Server | Web server (via XAMPP) |

### 2.3 Development Tools

• XAMPP: Local development environment (Apache + MySQL + PHP)
• phpMyAdmin: Database management interface
• Notepad: Code editor
• Chrome DevTools: Frontend debugging and testing

## 3. Database Design

The database consists of 15 tables storing all application data. Key tables include:

| Table Name | Description |
|---|---|
| users | User accounts and credentials |
| destinations | Tourist destinations information |
| tour_packages | Available tour packages |
| hotels | Hotel listings |
| bookings | User booking records |
| reviews | User reviews and ratings |
| provinces | Pakistani provinces |
| favorites | User favorite destinations |
| travel_tips | Travel guides and tips |
| contact_messages | User inquiries |

## 4. System Features

### 4.1 User Authentication

• User Registration: Create new account with email and password
• Login System: Secure authentication with password hashing
• Remember Me: 30-day automatic login via secure cookies
• Forgot Password: Email-based password recovery
• Session Management: Secure user sessions with timeout

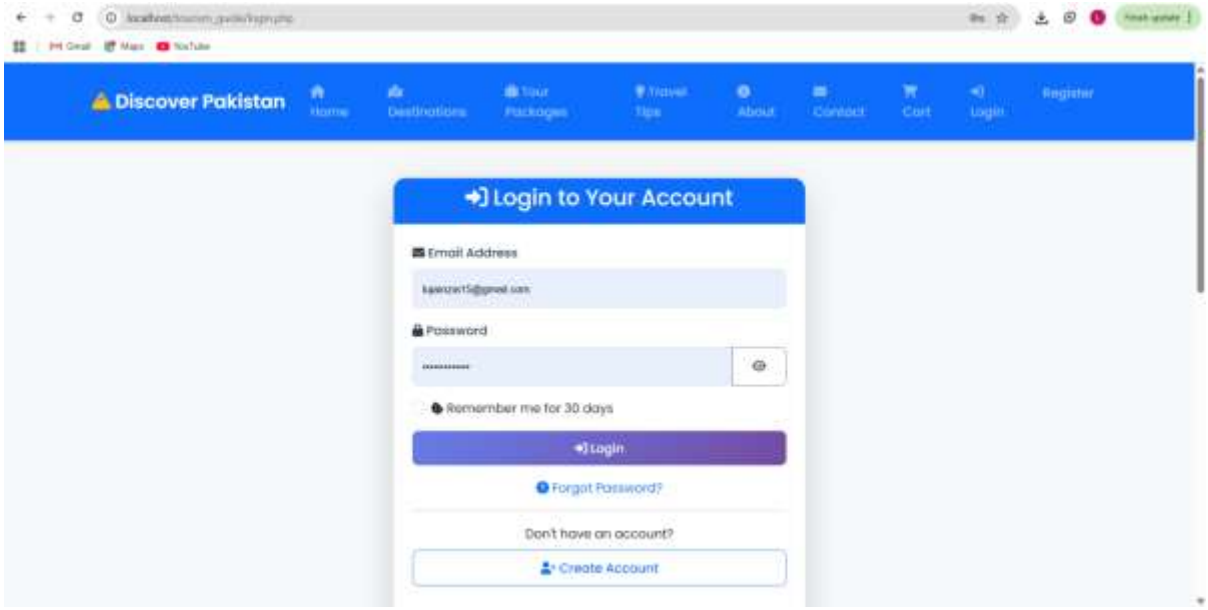



Figure 2: Login page.

## 4.2 Destination Browsing

• Browse Destinations: View all tourist destinations in Pakistan
• Filter Options: Filter by province, category, and rating
• Search Functionality: Search destinations by name
• Detailed View: See comprehensive information about each destination
• Featured Destinations: Highlighted popular locations

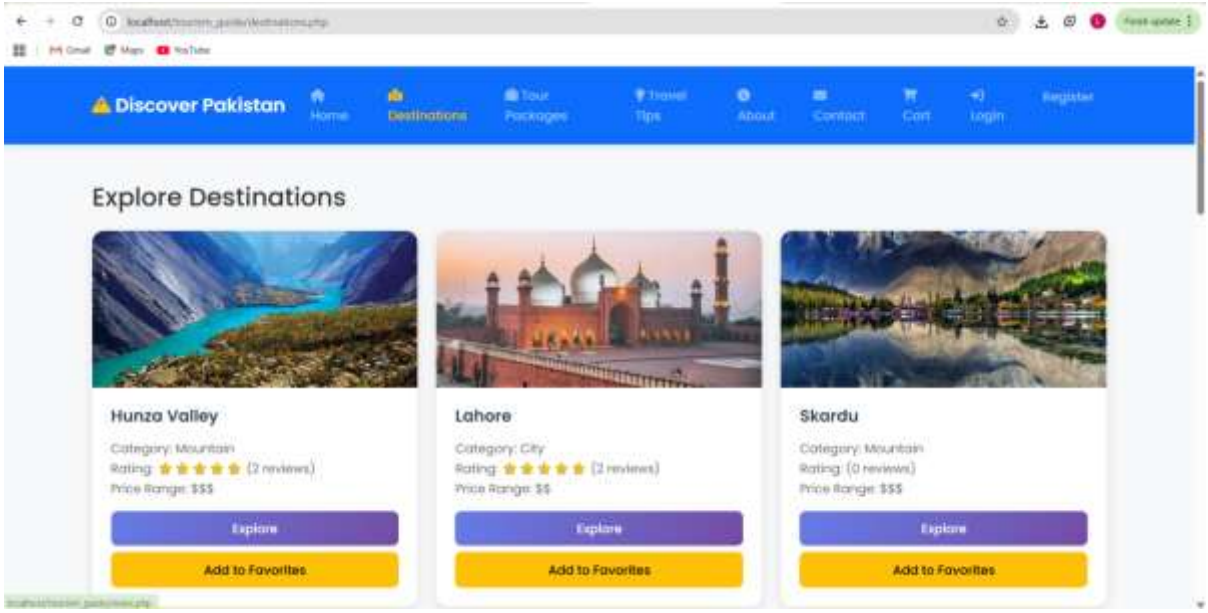



Figure 3: Destination page.

## 4.3 Tour Packages

• Package Listing: Browse available tour packages
• Package Details: View itinerary, pricing, and inclusions

• Duration & Difficulty: See trip length and difficulty level
• Price Calculator: Real-time price calculation based on guests
• Filter Packages: By province, price, and difficulty

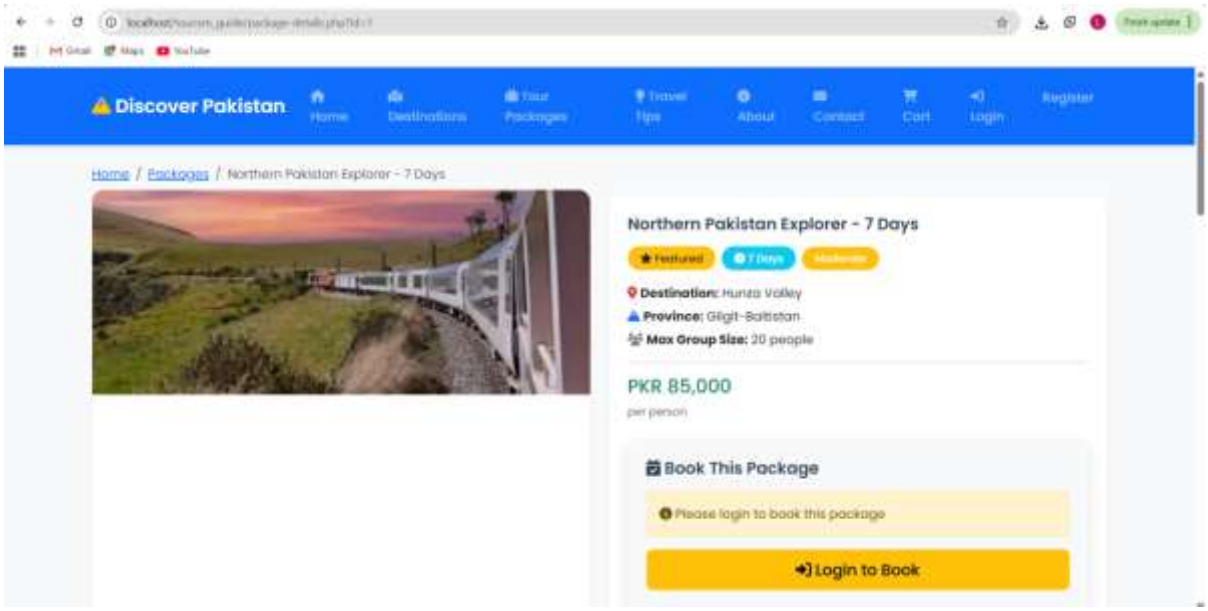



Figure 4: Package details.

## 4.4 Booking System

• Online Booking: Book tour packages directly online
• Date Selection: Choose check-in and check-out dates
• Guest Selection: Specify number of travelers
• Hotel Selection: Optional hotel booking
• Special Requests: Add custom requirements
• Booking Confirmation: Receive booking confirmation

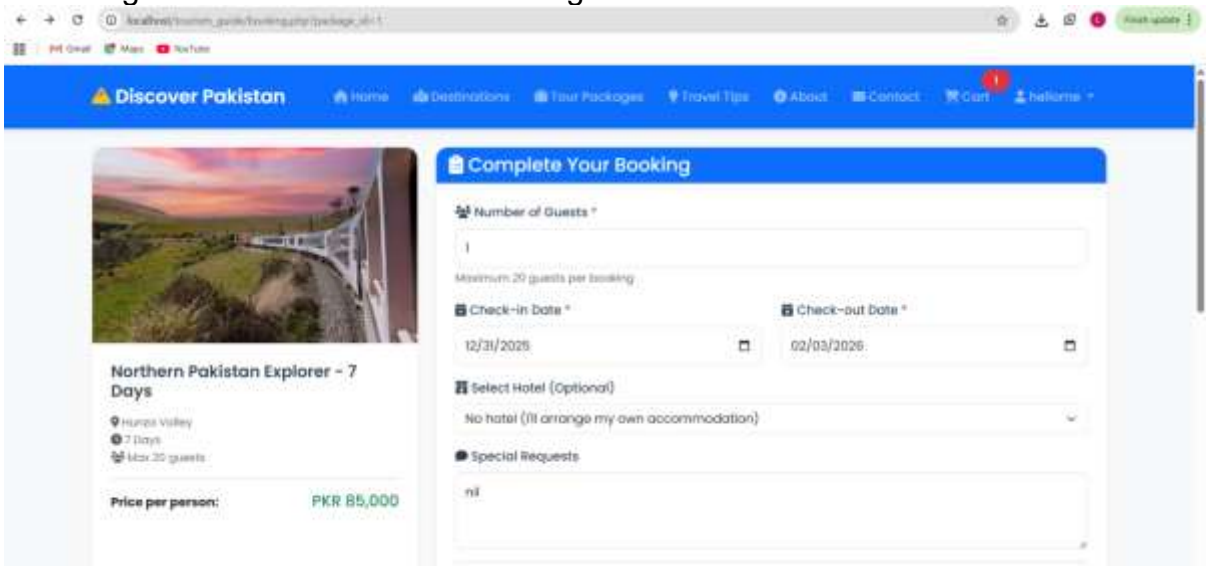



Figure 5: Booking page.

### 4.5 Shopping Cart

• Add to Cart: Add multiple packages before checkout
• Cart Management: View, update, or remove items
• Guest Count: Adjust number of guests per package
• Price Summary: View total cost with calculations
• Cart Persistence: Cart saved during browsing session

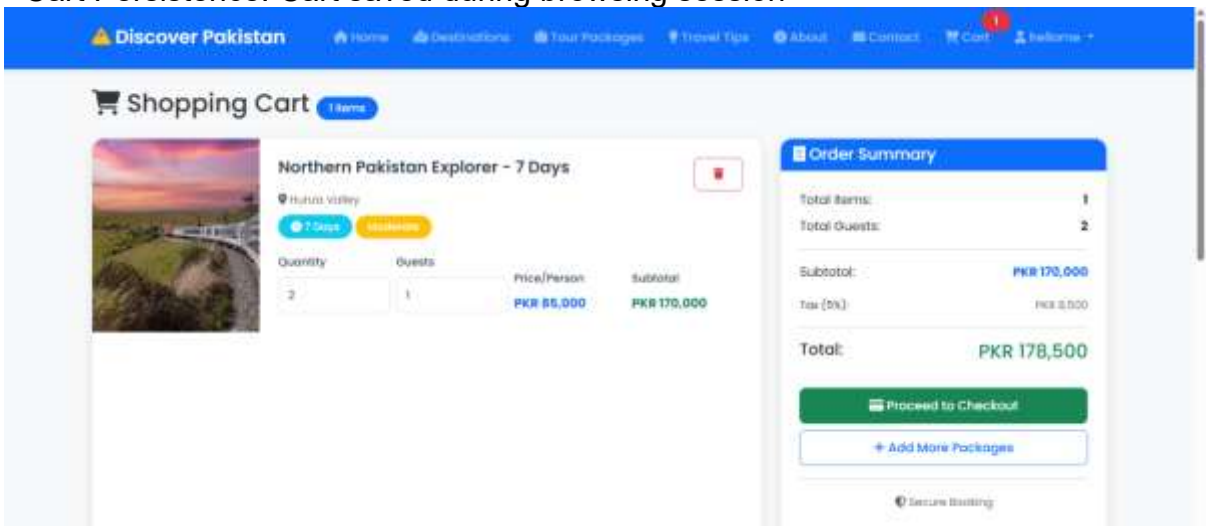



Figure 6: Shopping cart.

### 4.6 User Dashboard

• Profile Summary: View user information
• Booking History: See all past and active bookings
• Booking Status: Track booking status (pending, confirmed, completed)
• Quick Statistics: View total bookings and favorites

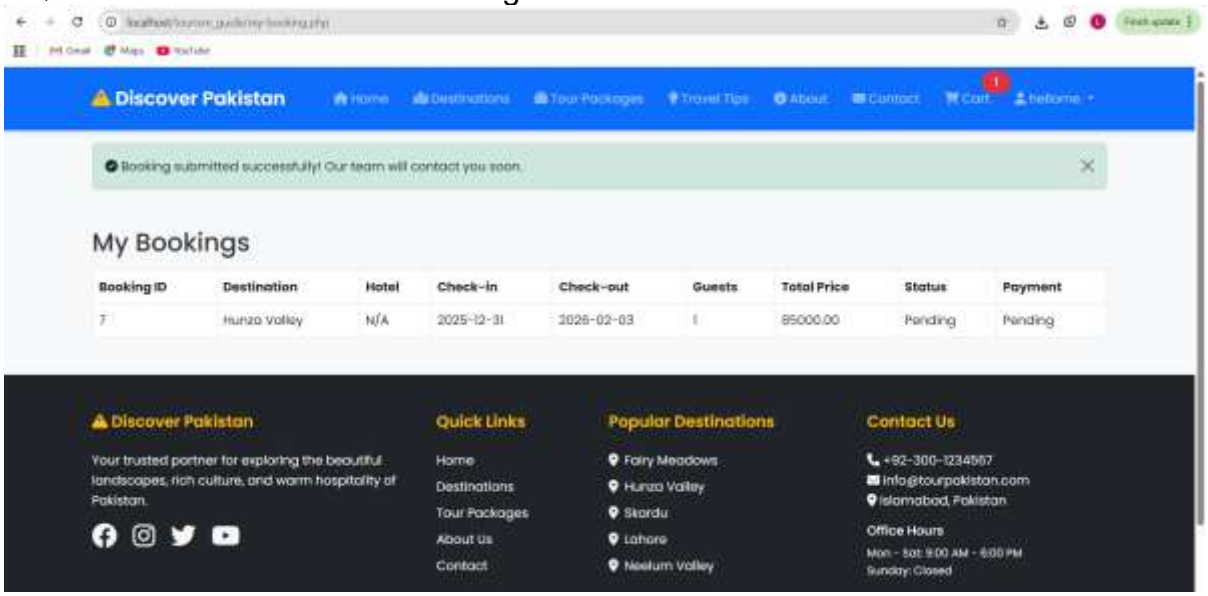



Figure 7: User dashboard.

### 4.7 Profile Management

• Change Password: Secure password update

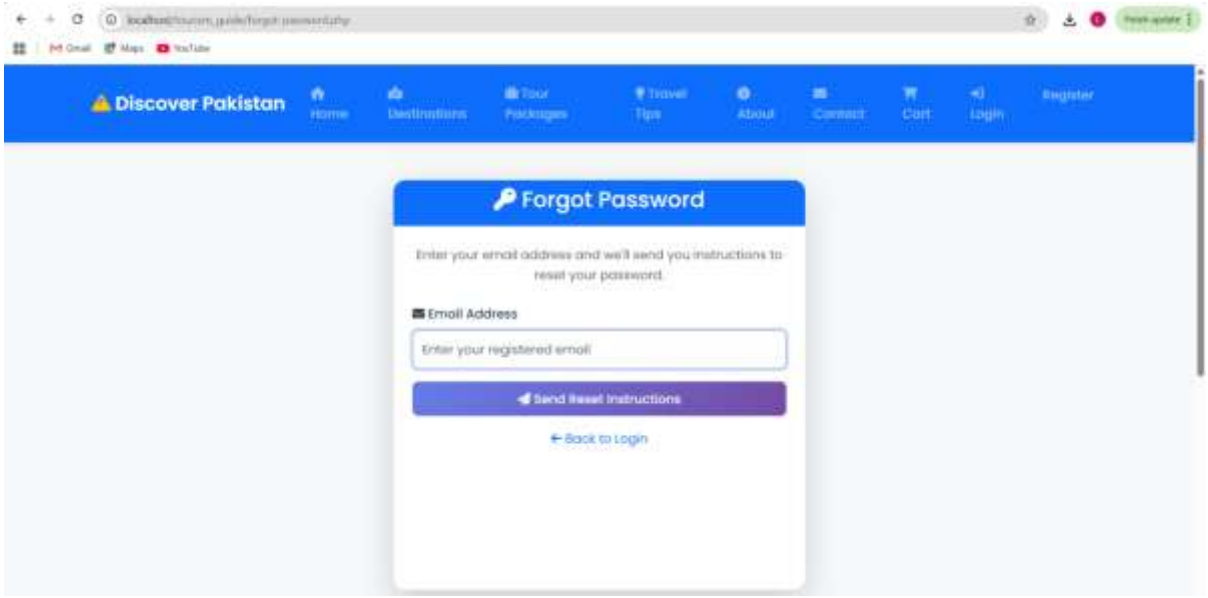



Figure 8: Forgot password page.

### 4.8 Additional Features

• Reviews & Ratings: Users can rate and review destinations
• Favorites: Save favorite destinations for later
• Travel Tips: Read helpful travel guides and tips
• Contact Form: Send inquiries to support team
• Responsive Design: Works on desktop, tablet, and mobile




Figure 9: Add a review.

### 4.9 Upload profile picture:

- Uploads profile picture.
- Tells what format image to upload.
- Tells maximum file size is 5MB.

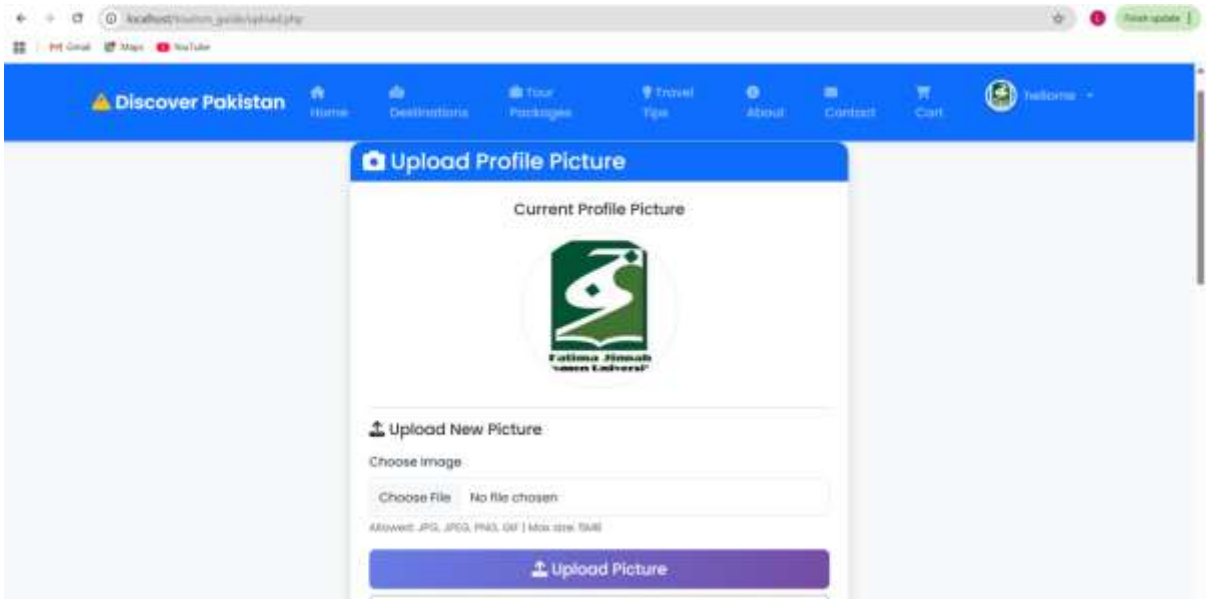



Figure 10: Uploading profile picture.

### 4.10 Admin functions

- After login, admin dashboard.

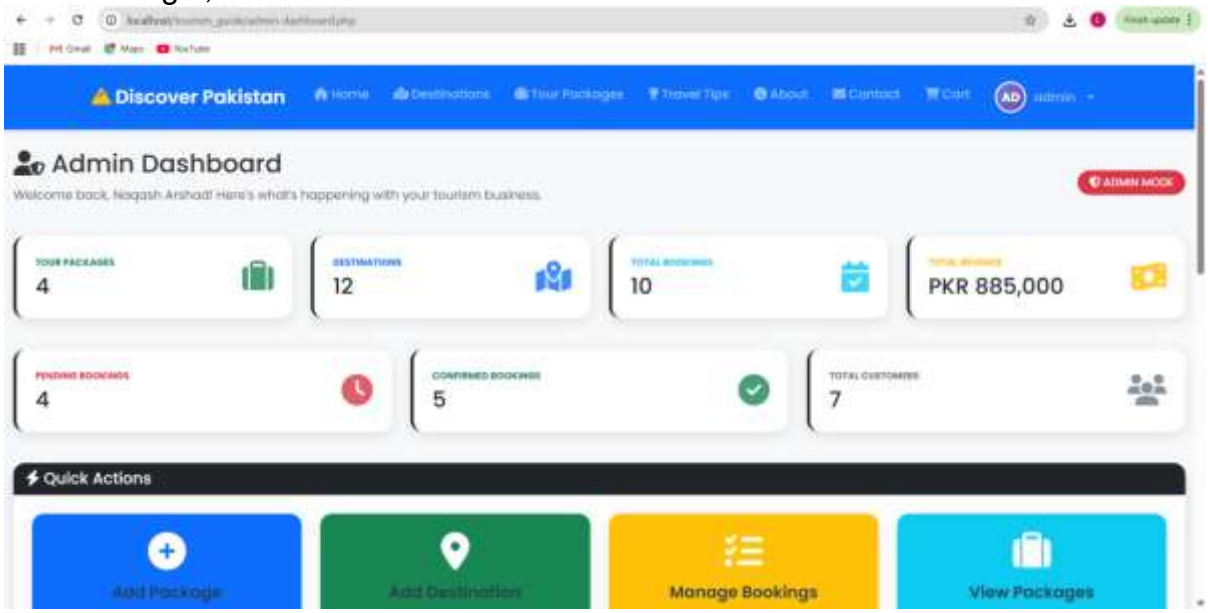



Figure 11: Admin Dashboard.

- Admin can delete, add manage bookings.

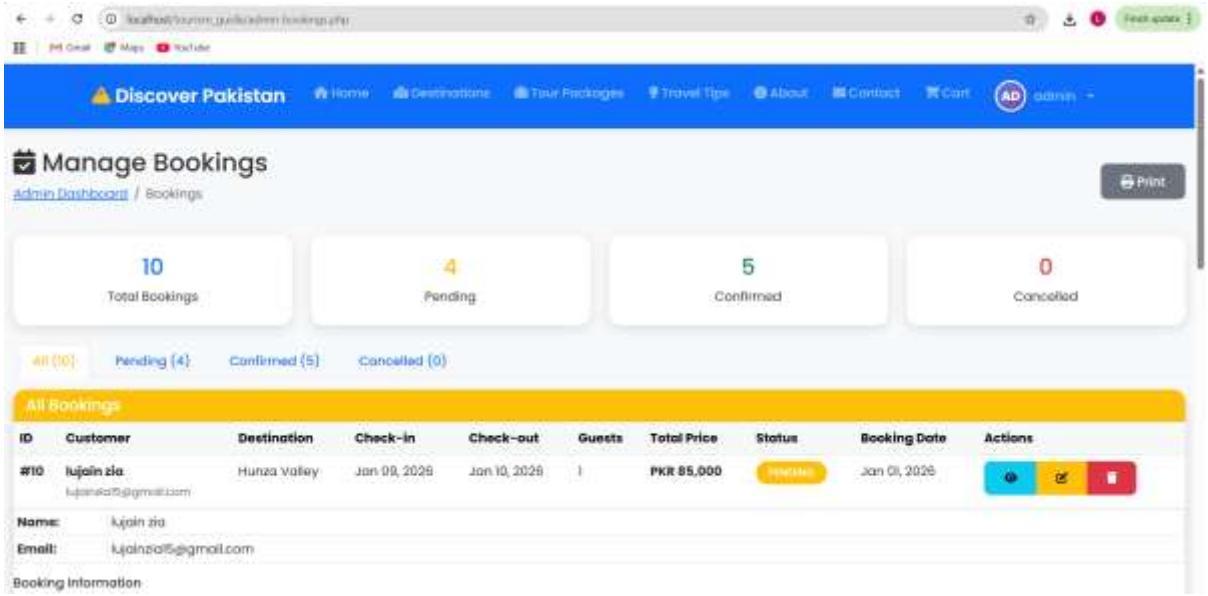



Figure 12 : Manage bookings.

- Admin can add, delete  packages.

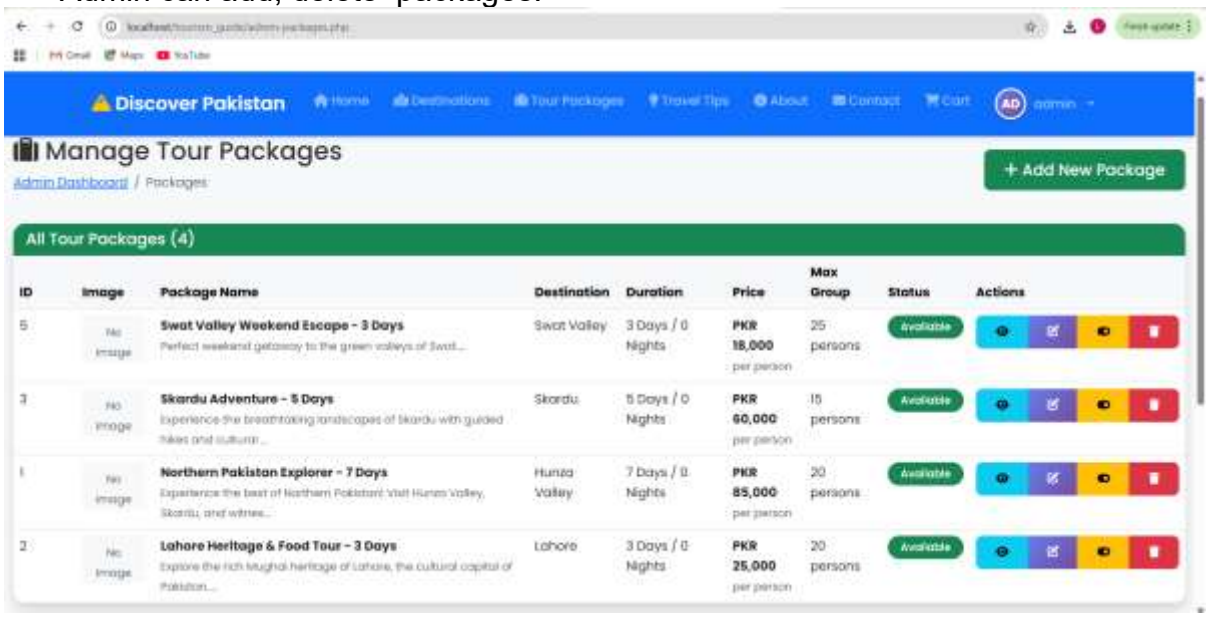



Figure 13: manage packages.

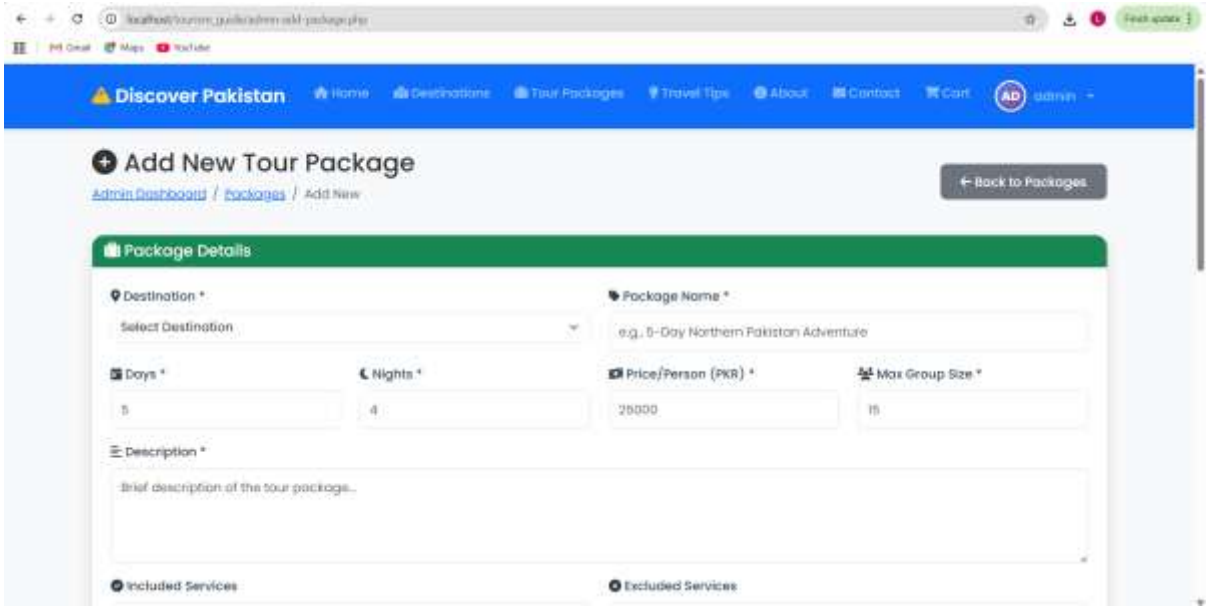



Figure 14: add package.

- Admin can add destinations.

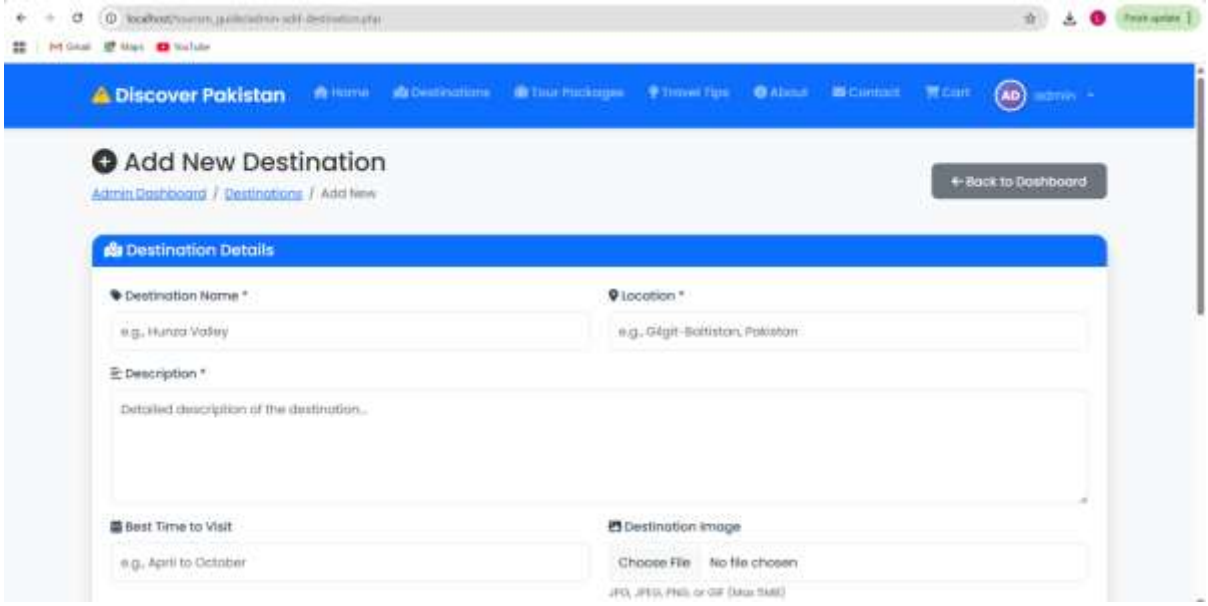



Figure 15: add destination.

## 5. How The System Works

### 5.1 User Registration & Login

1. User visits the website and clicks 'Register'
2. Fills in registration form with name, email, and password
3. System validates input and creates account
4. Password is hashed using bcrypt for security
5. User can login with email and password
6. System creates secure session for logged-in user
7. Optional 'Remember Me' creates 30-day cookie

### 5.2 Browsing & Selecting Packages

1. User browses destinations or packages
2. Can filter by province, price, or difficulty
3. Clicks on package to view full details
4. Reviews itinerary, pricing, and inclusions
5. Selects number of guests using counter
6. Price automatically calculates (guests × price per person)

### 5.3 Booking Process

1. User clicks 'Book Now' or 'Add to Cart'
2. If 'Book Now': Redirects to booking form
3. If 'Add to Cart': Package added to shopping cart
4. In booking form, user selects:
   - Check-in and check-out dates
   - Number of guests
   - Optional hotel
   - Special requests
5. System validates all inputs
6. Calculates total price
7. User confirms booking
8. Booking saved to database with 'pending' status
9. User can view booking in 'My Bookings' page

### 5.4 Shopping Cart Workflow

1. User adds packages to cart from any page
2. Cart stored in PHP session (temporary storage)
3. Cart icon shows item count badge
4. User can view cart anytime
5. In cart page, user can:
   - Update guest count
   - Remove items
   - Clear entire cart
6. Subtotal and tax calculated automatically
7. User proceeds to checkout
8. Each cart item converted to individual booking

### 5.5 Session Management

1. When user logs in, PHP session is created
2. Session stores: user_id, username, email, login_time
3. Session valid for 1 hour of inactivity
4. Session ID regenerated every 30 minutes for security
5. Shopping cart stored in session
6. On logout, session destroyed and cart cleared

## 6. Security Features

### 6.1 Password Security

• Passwords hashed using PHP password_hash() function
• Bcrypt algorithm with automatic salt generation

• Minimum password length requirement (6 characters)

### 6.2 SQL Injection Prevention

• Input sanitization using mysqli_real_escape_string()
• Input validation before database queries
• Type casting for numeric inputs

### 6.3 XSS Prevention

• Output escaping using htmlspecialchars()
• All user-generated content sanitized before display

### 6.4 Session Security

• HttpOnly flag set on session cookies
• Session ID regeneration every 30 minutes
• Automatic session timeout after 1 hour
• Session validation on every protected page

### 6.5 File Upload Security

• MIME type validation (not just file extension)
• File size restrictions (5MB maximum)
• Unique filename generation to prevent overwrites
• Upload directory protected with .htaccess
• Image re-encoding to remove malicious code

## 7. Website Pages

| Page Name | Description |
|---|---|
| index.php | Homepage with featured destinations and packages |
| login.php | User login page |
| register.php | New user registration |
| forgot-password.php | Password recovery |
| destinations.php | Browse all destinations |
| destination-details.php | Detailed destination information |
| packages.php | Browse tour packages |
| package-details.php | Detailed package information |
| booking.php | Booking form |
| cart.php | Shopping cart |
| dashboard.php | User dashboard |
| profile.php | Profile management |
| my-booking.php | View booking history |
| favorites.php | User's favorite destinations |
| hotels.php | Browse hotels |
| travel-tips.php | Travel guides and tips |
| contact.php | Contact form |
| about.php | About the website |
| upload.php | Form for profile picture upload |
| Upload-process.php | Contains info about uploading process |

## 8. Conclusion

The Pakistan Tourism Guide Website successfully demonstrates a complete web application with modern features and secure implementation. The project covers all essential aspects of web development including frontend design, backend programming, database management, and security implementation.

The system provides a user-friendly platform for exploring Pakistani tourism destinations and booking tour packages. With features like session management, shopping cart, file uploads, and secure authentication, the project showcases practical application of web development concepts.

### Key Achievements

- Developed 26 functional PHP pages
- Created 17-table normalized database
- Implemented secure user authentication
- Built responsive design for all devices
- Added shopping cart functionality
- Included file upload with image processing
- Applied security best practices

### Future Enhancements

- Real payment gateway integration (JazzCash, EasyPaisa)
- Live chat support
- Mobile application (iOS/Android)
- Real social media login (Google, Facebook)
- Multilingual support (Urdu)